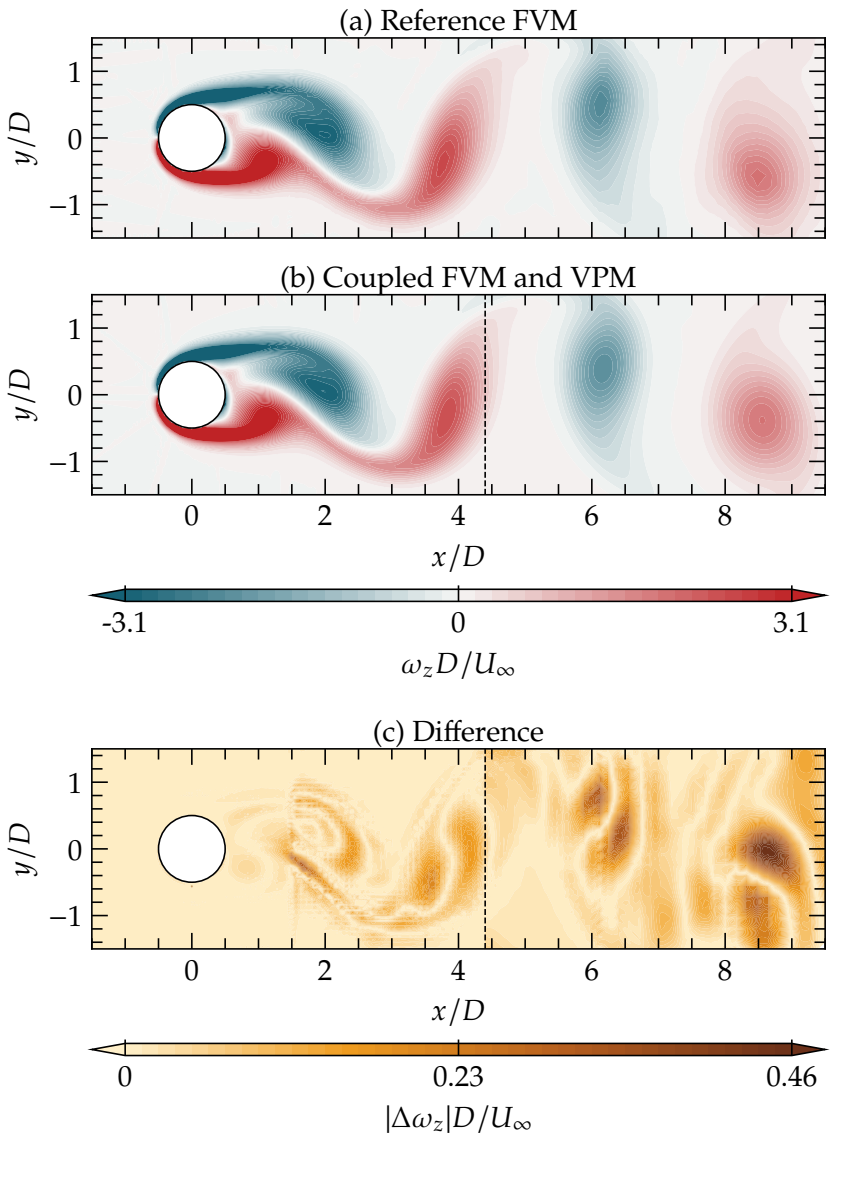

(a) Reference FVM
y/D
1
0
−1
(b) Coupled FVM and VPM
y/D
1
0
−1
0
2
4
6
8
x/D
-3.1
0
3.1
ω_z D/U_∞
(c) Difference
y/D
1
0
−1
0
2
4
6
8
x/D
0
0.23
0.46
|Δω_z|D/U_∞